

From Compliance to Decision Confidence: A Scoping Review of Accessible E-Commerce for Blind and Low-Vision People

Bektur Ryskeldiev
Mercari R4D
Tokyo, Japan
Research and Development Center for
Digital Nature
University of Tsukuba
Tsukuba, Japan
bektour@mercari.com

Matthew Gillingham
Mercari R4D
Tokyo, Japan
matthew.gillingham@mercari.com

Norimasa Kobori
Mercari R4D
Tokyo, Tokyo, Japan
n-kobori@mercari.com

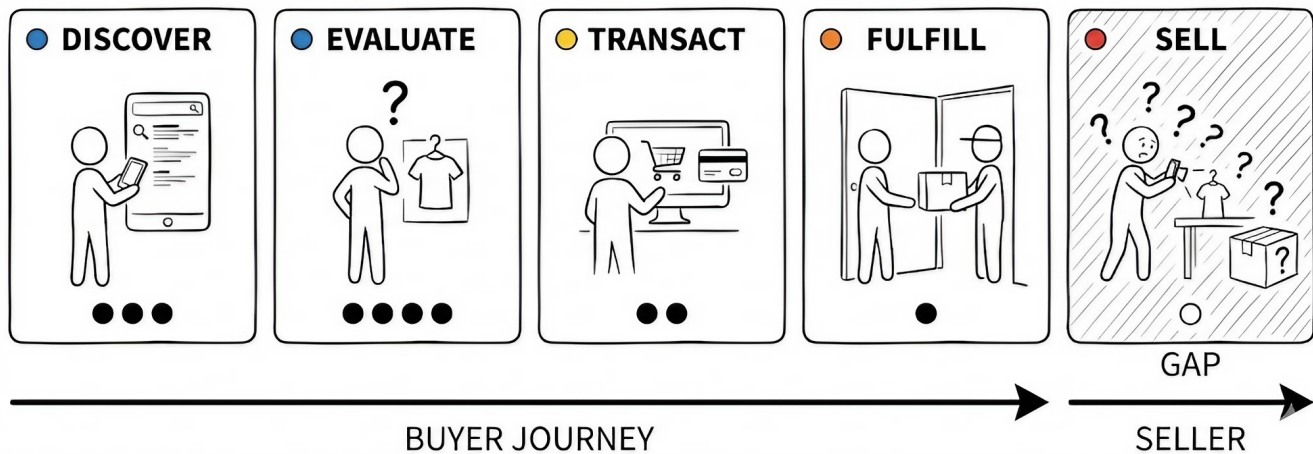


Figure 1: The e-commerce journey for BLV users spans five stages. Research concentrates on buyer-side activities (panels 1–4, indicated by filled dots showing research density), while seller workflows remain under-studied (panel 5, marked “GAP” with diagonal hatching). This scoping review maps the asymmetry and identifies seller-side accessibility as a critical research priority.

Abstract

Accessible e-commerce hinges on *decision confidence*: judging product suitability and transaction risk well enough to buy or sell without ad hoc sighted assistance. We present a scoping review synthesizing 21 studies (14 direct, 7 adjacent) on accessibility for blind and low-vision (BLV) shoppers and sellers. We derive four challenge clusters (visual content, navigation, fulfillment, trust) mapped onto five research approaches. Key gaps emerge: seller-side workflows are rarely supported by technical interventions (only 1/14 direct papers presents a seller-focused system), GenAI remains untested for seller tasks like product photography, and trust verification relies on user strategies rather than systematized tools. Excluding BLV people from selling limits entrepreneurship and gig economy

participation. We contribute a thematic framework, a challenge–approach matrix, and a research agenda.

CCS Concepts

• **Human-centered computing** → **Accessibility**; *Natural language interfaces*; • **Information systems** → **Online shopping**.

Keywords

accessibility, blind and low-vision, e-commerce, online shopping, scoping review, screen readers, generative AI



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

CHI EA '26, Barcelona, Spain

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799042>

ACM Reference Format:

Bektur Ryskeldiev, Matthew Gillingham, and Norimasa Kobori. 2026. From Compliance to Decision Confidence: A Scoping Review of Accessible E-Commerce for Blind and Low-Vision People. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799042>

1 Introduction

Online shopping now mediates activities from getting groceries to gig work; survey data show over 70% of blind users shop online frequently [20]. Yet e-commerce interfaces remain difficult despite two decades of work on screen-reader strategies [7] and accessibility standards [10, 36]. Automated audits flag missing labels and contrast problems [35]; a 2025 audit of leading e-commerce sites found 84% of checkout pages functionally invisible to screen readers [11], and on C2C marketplaces such as eBay, Etsy, and Mercari, seller-generated content is rarely screen-reader compatible. Yet these audits do not capture whether a person can discover, evaluate, purchase, and resell products independently. We use *decision confidence* to denote whether BLV people can judge product suitability and transaction risk well enough to complete a purchase or sale—without ad hoc sighted assistance. Qualitative studies confirm that BLV shoppers spend significant effort “debugging” page structure, guessing at visual affordances, and relying on sighted help [39].

Within HCI and accessibility research, a growing set of studies addresses e-commerce scenarios: shopping assistants combining computer vision and crowd support [27], redesigned search navigation for retail sites [34], and automated image descriptions [37]. Work on customer-to-customer (C2C) marketplaces reveals how peer-generated photos and reputation systems create distinct challenges for both BLV buyers and sellers [23, 24]. Adjacent work on voice assistants [21, 22] suggests these interfaces have accessibility potential, though shopping-specific research remains sparse.

Outside explicitly commerce-focused systems, BLV people adopt general-purpose generative AI tools as “meta” accessibility layers that interpret complex screens and images across tasks, including shopping [1]. Recent studies characterize this as “everyday uncertainty”: a mindset of skepticism toward both AI- and human-mediated assistance [28]. This uncertainty can erode decision confidence, as users may hesitate to rely on AI-mediated product descriptions without verification mechanisms. Industry deployments such as Be My AI [4] show growing adoption, yet WCAG conformance does not guarantee usable shopping [32, 33]. Regulatory pressure mounts (European Accessibility Act [12]), but compliance frameworks measure code violations, not decision confidence.

This poster presents a scoping review addressing three questions: (1) What recurring barriers do BLV people face across the e-commerce journey? (2) What technical and empirical approaches has research employed? (3) Where are the gaps—especially for sellers and trust verification? To our knowledge, no prior synthesis covers BLV end-to-end e-commerce including seller workflows. We synthesize 21 papers from HCI, accessibility, and adjacent venues. Contributions: (1) a thematic framework of four challenge clusters (visual content, navigation, fulfillment, trust); (2) a challenge–approach matrix mapping clusters onto five research approaches; (3) a research agenda prioritizing seller-side accessibility and end-to-end evaluation. Figure 1 shows the framework.

2 Scope and Methods

2.1 Scope and inclusion criteria

We focus on end-to-end e-commerce experiences of BLV people as both buyers and sellers. We include two types of work: (1) **direct e-commerce studies** where shopping or selling is a primary task

context, and (2) **adjacent studies** that involve BLV users and study capabilities directly mediating at least one e-commerce stage (e.g., image descriptions for product evaluation, authentication during checkout, remote assistance for verifying delivered goods). We exclude general web accessibility studies, payment/banking-only work, and studies not involving BLV users.

2.2 Search and selection

We searched ACM DL, Google Scholar, and Springer/Elsevier databases from 2010–2025 (search conducted January 2026). For ACM DL, we used: `[[All: “online shopping”] OR [All: “e-commerce”] OR [All: “marketplace”]] AND [[All: “blind”] OR [All: “low vision”] OR [All: “screen reader”] OR [All: “visual impairment”]]`. We adapted equivalent queries for other databases and supplemented with backward/forward snowballing. Initial search returned 142 candidates; 115 were excluded at title/abstract screening (not BLV-focused, $n=52$; not e-commerce-related, $n=41$; duplicates, $n=22$). Of 27 full texts reviewed, 6 were excluded (general web accessibility only, $n=4$; no user involvement, $n=2$), yielding 21 included papers.

Table 1: Corpus breakdown by study type.

Type	Direct (N=14)	Adjacent (N=7)
Retail/nav	Page restructuring [27, 34]	Image descriptions [26, 37]
Fashion	Clothing barriers [15]	Voice assistants [21, 22]
Mobile	Shopping apps [16]	Remote assistance [3]
Compliance	WCAG studies [32]	Authentication [2]
C2C	Marketplaces [23, 24, 30]	Service trust [9]
Qualitative	Survey, GenAI use [1, 20, 29, 39]	—
Fulfillment	Parcel/shipping [17, 25]	—

We also cite emerging work on BLV photography privacy [14] and GenAI page remediation [38].

2.3 Coding

For each paper we extracted scenario, participants, methods, and findings. We coded challenge passages into four clusters (visual content, navigation, fulfillment, trust) and tagged papers with approach families (GenAI/LLMs, computer vision, wearables/telepresence, audits/guidelines, qualitative studies). This scoping review employs reflexive thematic analysis [8], where the researcher’s interpretive role is acknowledged rather than minimized. The first author coded all papers; codes were applied with memoing to document boundary cases and rationale. LLMs were used only for writing support and to prompt alternative framings during memoing (e.g., “What alternative groupings exist?”); they did not perform screening or coding. We invite community refinement at the poster session.

3 Findings: Challenges and Approaches

Across 21 studies (14 direct e-commerce, 7 adjacent) we identify four recurring **challenge clusters** and five **research approaches**. Adjacent studies on image descriptions, voice assistants, remote

assistance, authentication, and service fulfillment trust inform the design space but are not shopping-specific.

Corpus summary: Of 14 direct e-commerce papers, 10 address buyer tasks, 5 address seller tasks (some address both), and 3 propose seller-side technical interventions.

3.1 Challenge clusters

Visual content access and aesthetics. Product photography conveys fit, color, material, and style—information that alt text rarely captures [15, 39]. Several systems address this gap: BrowseWithMe and Revamp [27, 34] restructure product imagery for screen readers, while AI-generated descriptions [37] automate access at scale. However, BLV users want context-sensitive detail rather than generic labels [26], and when platform descriptions lack nuance, users resort to external tools for verification. This gap between available and needed visual information directly undermines decision confidence.

Navigation and information architecture. E-commerce sites present deep hierarchies and inconsistent layouts that force BLV shoppers into effortful browsing strategies [7, 15]. Product grids and carousels, designed for visual scanning, become long linear reading orders under a screen reader. Mobile apps with layout restructuring [16] show promise, but most work treats navigation as a page-level problem rather than as a flow spanning discovery through post-purchase. The resulting cognitive load fragments the shopping journey and erodes decision confidence.

Physical handling and fulfillment. Shopping and selling involve material labor—listing, photographing, packaging, shipping—that extends well beyond the screen [23, 30]. Remote assistance helps verify labels and barcodes [3], while work on parcel lockers [25] and end-to-end shopping journeys [17] confirms that fulfillment accessibility matters. Similar barriers appear in adjacent domains like food delivery [31]. Notably, seller-side fulfillment remains a distinct and under-explored design space, where decision confidence failures cascade into transaction abandonment or reliance on sighted intermediaries.

Trust and verification. BLV participants consistently describe a gap between retailer claims and their own purchase confidence [1, 23, 39]. To bridge this gap, people combine platform reviews, remote sighted assistance (e.g., Be My Eyes [3]), and general-purpose GenAI tools (e.g., Be My AI, ChatGPT) to cross-check product information [1, 29]. Additional friction arises from inaccessible authentication [2] and voice interfaces that surface too few options [21, 22]. Trust dynamics in service fulfillment [9] parallel those in marketplace transactions. These barriers directly erode decision confidence, measurable through proxies like time-to-confidence, reliance on sighted help, and cart abandonment.

3.2 Approach families and challenge–approach matrix

Key gaps: Only 1/14 direct papers proposes a seller-side technical system [30]; two others investigate seller-side barriers [13] or fulfillment accessibility [25]. The Fulfillment row gaps (Table 2) confirm the community treats e-commerce as digital-only. GenAI addresses buyer tasks (1 paper); no papers in our corpus address

Table 2: Challenge–Approach Matrix showing corpus papers (N=21). Empty cells indicate research gaps; shaded cells mark critical gaps with no technical interventions.

	GenAI	CV	Wearable	Audits	Qual.
Visual	[1]	[27, 34, 37]	[3]	—	[15, 26, 39]
Nav.	—	[27, 34]	[16]	[32]	[15]
Fulfill.	—	—	[3, 25]	—	[17, 23, 30]
Trust	[1]	—	[3, 21, 22]	[2]	[9, 23, 29, 39]

seller workflows with GenAI. Trust relies on qualitative strategies (4 papers); automated verification is absent.

4 Research Agenda

Based on our synthesis, we identify three priorities for future research.

4.1 From point fixes to structural remediation

Most interventions target specific UI components (image descriptions, filters). One promising direction is generative AI for structural remediation: fixing document semantics, summarizing screens, exposing cross-page relationships. For example, AI could transform a 15-image product gallery into a structured summary: “3 lifestyle shots, 2 size charts, 10 product angles; alt-text missing on 6.” Recent work characterizes how BLV users approach AI-mediated information access with “everyday uncertainty”—maintaining skepticism while still deriving value [28]. Be My AI and InnoSearch [4, 6] show AI accessibility tools reaching mainstream use. [1]; future work should develop techniques for inspecting AI-mediated changes, debugging errors, and calibrating trust for high-stakes purchase decisions. Beyond fixing existing pages, AI agents may eventually act on behalf of users—browsing, comparing, and purchasing autonomously—reducing reliance on accessible interfaces altogether. These opportunities require careful evaluation due to hallucination and error risks in e-commerce contexts.

4.2 Centering BLV sellers and workers

C2C marketplace studies show accessibility challenges extend to BLV sellers, gig workers, and small business owners [23]. One seller reported spending three hours listing a single item, outsourcing photo verification via Be My Eyes [30]—a decision-confidence failure cascading from buyer tools to seller burden. Fulfillment tasks (measuring, labeling, tracking parcels) present real obstacles [13]. Future tools could support listing workflows: AI that generates product descriptions from seller narration, accessible photography guidance, and inventory management interfaces. Wearables may help [5], but integration into seller workflows needs study. Seller-side accessibility also shapes buyer decision confidence: sellers control much of the evidence buyers rely on (photos, condition notes, shipping promises). With C2C platforms and gig economy participation growing, excluding BLV people from selling limits economic independence.

4.3 End-to-end evaluation and decision confidence metrics

Future evaluation frameworks could move beyond page-level audits to measure *decision confidence*: can users trust that product attributes match reality, transactions complete without hidden costs, and seller verification is accurate? Decision confidence is well-studied in judgment and decision-making research [18]; some challenges (e.g., assessing price fairness or seller honesty) affect all users, but become acute for BLV people who lack visual verification channels. Measurable outcomes (time-to-confidence, reliance on sighted help, cart abandonment) differ from WCAG conformance. The European Accessibility Act [12] applies to e-commerce services from June 2025, yet compliance still does not ensure usability. Longitudinal field studies should follow BLV shoppers and sellers across platforms, complementing automated monitors [10, 19, 36] with task-based metrics.

4.4 Broader disability perspectives

Our review focuses on BLV users, but structured product information, simplified navigation, and multimodal descriptions may also benefit neurodivergent users and people with cognitive disabilities. Future work should examine how accessible e-commerce interventions transfer across disability communities.

5 Limitations

This poster-length scoping review has limitations. The corpus of 21 papers reflects a nascent BLV e-commerce literature; relevant work in payments, logistics, and non-HCI venues may be missing. A broader survey incorporating commercial accessibility audits of major platforms (e.g., Amazon, Shopify, eBay) alongside academic literature could yield a more comprehensive picture. Because our search emphasized English-language publications and major digital libraries, the synthesis may under-represent non-English scholarship and region-specific platforms. Coding was conducted by a single researcher; while reflexive thematic analysis foregrounds interpretive synthesis, confirmability can be strengthened in future work through independent auditing or parallel coding of a subset of papers. We mitigate this by inviting community critique at the poster session. Accordingly, our findings should be read as a mapping of themes and gaps rather than a complete assessment of e-commerce accessibility in practice.

Use of AI Writing Assistance

This extended abstract was drafted with the assistance of large language models (Claude Opus 4.5, Gemini 3 Pro, GPT 5.2-Pro). LLMs transformed author-prepared notes and coded data into prose, suggested restructuring, and served as critical reviewers. The authors defined research questions, conducted the literature search and coding, and are responsible for all final content.

References

- [1] Rudaiba Adnin and Maitraye Das. 2024. "I look at it as the king of knowledge": How Blind People Use and Understand Generative AI Tools. In *ASSETS '24*. doi:10.1145/3663548.3675631
- [2] S. Andrew, S. Watson, T. Oh, and G. W. Tigwell. 2020. A Review of Literature on Accessibility and Authentication Techniques. In *ASSETS '20*. doi:10.1145/3373625.3418005 Authentication friction in e-commerce; login/payment barriers.
- [3] Mauro Avila, Katrin Wolf, Anke Brock, and Niels Henze. 2016. Remote Assistance for Blind Users in Daily Life: A Survey about Be My Eyes. In *PETRA '16*. doi:10.1145/2910674.2935839
- [4] Be My Eyes. 2023. Be My AI: GPT-4 Powered Virtual Volunteer for Blind and Low-Vision Users. <https://www.bemyeyes.com/blog/announcing-be-my-ai>
- [5] Be My Eyes. 2024. Be My Eyes and Meta Announce Accessibility Partnership for Ray-Ban Meta Glasses. <https://www.bemyeyes.com/news/be-my-eyes-and-meta-announce-accessibility-partnership/>
- [6] Be My Eyes and InnoSearch. 2024. In-App Accessible Shopping for Blind and Low Vision Users. <https://www.bemyeyes.com/news/be-my-eyes-to-provide-in-app-accessible-shopping/>
- [7] Yevgen Borodin, Jeffrey P Bigham, Glenn Dausch, and IV Ramakrishnan. 2010. More than meets the eye: a survey of screen-reader browsing strategies. In *W4A '10*. doi:10.1145/1805986.1806005
- [8] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health* 11, 4 (2019), 589–597. doi:10.1080/2159676X.2019.1628806 Argues IRR is theoretically incoherent for reflexive TA; researcher's interpretive role is acknowledged.
- [9] Robin N. Brewer and Vaishnav Kameswaran. 2019. Understanding Trust, Transportation, and Accessibility through Ridesharing. In *CHI '19*. doi:10.1145/3290605.3300425 Service fulfillment trust; parallels marketplace seller trust.
- [10] World Wide Web Consortium. 2023. Web Content Accessibility Guidelines (WCAG) 2.2. <https://www.w3.org/TR/WCAG22/> Published October 5, 2023; adds 9 new success criteria.
- [11] Contentsquare Foundation. 2025. Ecommerce Accessibility: 2025 Readiness Snapshot. <https://contentsquare.com/press/e-commerce-accessibility-snapshot/> 84% of checkout pages functionally invisible to screen readers. Accessed February 2026.
- [12] European Commission. 2019. European Accessibility Act. <https://ec.europa.eu/social/main.jsp?catId=1202> Directive (EU) 2019/882; applicable from June 28, 2025. Accessed January 2026.
- [13] Kunal Gupta, Ken Takaki, Ayaka Tsutsui, Kentaro Teramoto, and Bektur Ryskeldiev. 2025. AccessibleShip: Exploring Packaging and Shipping Barriers for Visually Impaired Users in C2C E-Commerce. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8. doi:10.1145/3706599.3706665
- [14] Danna Gurari, Qing Li, Chi Chi, Yanan Zhao, Shuai Lin, Abigale Stangl, and Jeffrey P. Bigham. 2019. VizWiz-Priv: A Dataset for Recognizing the Presence and Purpose of Private Visual Information in Images Taken by Blind People. In *CVPR '19*. doi:10.1109/CVPR.2019.00961 Privacy risks when BLV sellers upload photos; supports seller-side argument.
- [15] Marylu Gutiérrez-López, Allison Woodruff, and Sarah Horton. 2023. Barriers to Clothing Websites for Visually Impaired People. In *DIS '23*. doi:10.1145/3563657.3595978 Focused study on online clothing site barriers; visual content + navigation clusters.
- [16] Min-Jeong Kim, Jae-Hyun Park, and Seung-Hyun Lee. 2020. BarrierFreeShop: Accessible Mobile Shopping Application for Blind Users. *International Journal of Advanced Smart Convergence (IJASC)* 9, 3 (2020), 260–271. doi:10.7236/IJASC.2020.9.3.260 Mobile shopping app with layout automation and review summarization; 80 participants.
- [17] Sooyeon Lee, Xiyang Wang, and John M. Carroll. 2020. Constructing a holistic view of shopping with people with visual impairment: a participatory design study. *Universal Access in the Information Society* 19 (2020), 469–483. doi:10.1007/s10209-018-0633-z End-to-end shopping journey; planning, navigating, unpacking.
- [18] Sarah Lichtenstein, Baruch Fischhoff, and Lawrence D. Phillips. 1982. Calibration of probabilities: The state of the art to 1980. In *Judgment Under Uncertainty: Heuristics and Biases*, Daniel Kahneman, Paul Slovic, and Amos Tversky (Eds.). Cambridge University Press, 306–334. doi:10.1017/CBO9780511809477.023
- [19] Mada Center. n.d. Mada Web Accessibility Monitor. <https://monitor.mada.org.qa/> Automated accessibility monitoring tool. Accessed January 2026.
- [20] Yoshiki Nagatani, Kazuki Takazawa, Masaaki Sadasue, Mariko Kobayashi, Kentaro Teramoto, Nozomu Yoshida, Kotaro Oomori, Akihiisa Shitara, Koyomi Kobayashi, Aiko Kuroda, and Misato Takahashi. 2022. How Visually Impaired People Use Information Media and E-Commerce in Japan. In *HCI International 2022 Posters (CCIS, Vol. 1580)*. Springer. doi:10.1007/978-3-031-06417-3_74 Survey of 133 BLV users; 70%+ blind shop online always/often; seller barriers data.
- [21] Alisha Pradhan, Leah Findlater, and Amanda Lazar. 2019. "Phantom Friend" or "Just a Box with Information": Personification and Ontological Categorization of Smart Speaker-Based Voice Assistants by Older Adults. In *CSCW '19*. doi:10.1145/3359316 Voice assistant use patterns relevant to commerce.
- [22] Alisha Pradhan, Kanika Mehta, and Leah Findlater. 2018. "Accessibility Came by Accident": Use of Voice-Controlled Intelligent Personal Assistants by People with Disabilities. In *CHI '18*. doi:10.1145/3173574.3174033 Foundational work on voice assistant accessibility.
- [23] Bektur Ryskeldiev, Kotaro Hara, Mariko Kobayashi, and Koki Kusano. 2022. Investigating Accessibility Challenges and Opportunities for Users with Low Vision Disabilities in Customer-to-Customer (C2C) Marketplaces. In *Proceedings of the*

- 24th International ACM SIGACCESS Conference on Computers and Accessibility. 1–4. doi:10.1145/3517428.3550390
- [24] Bektur Ryskeldiev, Mariko Kobayashi, Kentaro Teramoto, and Koki Kusano. 2023. Towards Immersive Inclusivity for C2C: How Immersive Multimodal Interactions Can Make Online Customer-to-Customer Shopping More Inclusive. In *Proceedings of the Asian HCI Symposium 2023*. 84–88. <https://dl.acm.org/doi/proceedings/10.1145/3574860>
- [25] Yuhang Shi, Mingming Fan, Xuhai Xu, Enhao Zhao, Toby Jia-Jun Trova, Jeffrey P. Bigham, and Amy Pavel Wang. 2024. FetchAid: Making Parcel Lockers More Accessible to Blind and Low Vision People With Deep-learning Enhanced Touchscreen Guidance, Error-Recovery Mechanism, and AR-based Search Support. In *CHI '24*. doi:10.1145/3613904.3642213 Fulfillment accessibility; parcel locker navigation for BLV users.
- [26] Abigale Stangl, Meredith Ringel Morris, and Danna Gurari. 2020. “Person, Shoes, Tree. Is the Person Naked?” What People with Vision Impairments Want in Image Descriptions. In *CHI '20*. doi:10.1145/3313831.3376404 User preferences for image descriptions; applicable to product images.
- [27] Abigale J Stangl, Esha Kothari, Suyog D Jain, Tom Yeh, Kristen Grauman, and Danna Gurari. 2018. BrowseWithMe: An Online Clothes Shopping Assistant for People with Visual Impairments. In *ASSETS '18*. 107–118. doi:10.1145/3234695.3236337
- [28] Lotus Tang, Ali Abdolrahmani, Darren Gergle, and Anne Marie Piper. 2025. Everyday Uncertainty: How Blind People Use GenAI Tools for Information Access. In *CHI '25*. doi:10.1145/3706598.3713433
- [29] Dhruv Thakkar, Harini Sampath, Sile O'Modhrain, and Shiri Azenkot. 2024. “We are at the mercy of others’ opinion”: Supporting Blind and Low Vision People in Recreational Online Shopping. In *DIS '24*. doi:10.1145/3677846.3677860 Recreational and window shopping experiences for BLV users.
- [30] Ayaka Tsutsui, Kengo Tanaka, Yaman Yu, and Bektur Ryskeldiev. 2025. “I Am a Blind Seller!”: Picture Taking Assistance for Visually Impaired Individuals for Participation as Sellers in Customer to Customer (C2C) Marketplaces. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–10. doi:10.1145/3706599.3720150
- [31] Pedro Vazquez, Marcos Baez, and Fabio Casati. 2023. Accessibility of Food Delivery Apps for Blind and Visually Impaired Users. In *MobileHCI '23 Companion*. doi:10.1145/3565066.3608688 Platform-based fulfillment + trust barriers in delivery context.
- [32] Beat Vollenwyder, Klaus Opwis, and Florian Brühlmann. 2023. Effects of web accessibility on user experience: Implications for online shopping. *International Journal of Human-Computer Studies* 170 (2023), 102956. doi:10.1016/j.ijhcs.2022.102956 RCT: participants with visual impairments on online shop with varying WCAG conformance.
- [33] Beat Vollenwyder, Klaus Opwis, and Florian Brühlmann. 2023. How compliance with web accessibility standards shapes the experiences of users with and without disabilities. *International Journal of Human-Computer Studies* 170 (2023), 102944. doi:10.1016/j.ijhcs.2022.102944
- [34] Ruolin Wang, Zixuan Chen, Mingrui Ray Zhang, Shan Lee, Suyeon Cho, Uran Oh, Leah Findlater, and Jeffrey P. Bigham. 2021. Revamp: Enhancing Accessible Information Seeking Experience of Online Shopping for Blind or Low Vision Users. In *CHI '21*. doi:10.1145/3411764.3445547
- [35] WebAIM. 2025. The WebAIM Million: The 2025 Report on the Accessibility of the Top 1,000,000 Home Pages. <https://webaim.org/projects/million/> 95% of pages had WCAG failures; 51 errors per page average. Accessed January 2026.
- [36] World Wide Web Consortium. n.d.. Web Content Accessibility Guidelines (WCAG) Overview. <https://www.w3.org/WAI/standards-guidelines/wcag/> Accessed January 2026.
- [37] Shaomei Wu, Jeffrey Wieland, Omid Farivar, and Julie Schiller. 2017. Automatic Alt-text: Computer-generated Image Descriptions for Blind Users on a Social Network Service. In *CSCW '17*. doi:10.1145/2998181.2998364 Platform-scale automated image descriptions.
- [38] Yaman Yu, Bektur Ryskeldiev, Ayaka Tsutsui, Michael Gillingham, and Yang Wang. 2025. From Cluttered to Clear: Improving the Web Accessibility Design for Screen Reader Users in E-commerce With Generative AI. In *ASSETS '25*. doi:10.1145/3663548.3675611 GenAI-based structural remediation of e-commerce pages.
- [39] Chuyi Yuan, Dihua Bi, Xiaosong Yeung, Shenjie Gao, Zhiliang Ye, and Jingyi Lu. 2019. “I Bought This for Me to Look More Ordinary”: A Study of Blind People Doing Online Shopping. In *CHI '19*. doi:10.1145/3290605.3300841 Qualitative study of how blind people shop online.